# DESMOPRESSIN

Hyponatraemia — fluid restriction

Nocturnal enuresis

**CHEMICAL NAME** 1-(3-mercaptopropanoic acid)-8-D-arginine vasopressin (DDAVP), acetate

**CLASS** Synthetic vasopressin (ADH) analogue — V2 agonist

**TRADE NAMES** DDAVP · Minirin · Octostim · Nocdurna

## THERAPEUTIC USES / INDICATIONS

- Cranial (central) diabetes insipidus.
- Primary nocturnal enuresis in children ≥5 yr.
- Nocturia due to nocturnal polyuria (adults).
- Mild haemophilia A and von Willebrand disease type 1 (↑ factor VIII/vWF) — IV/nasal; diagnostic tests (renal concentrating capacity).

## ADMINISTRATION

- **PO** tablets, **sublingual** melts, **intranasal** spray, **IV/SC**.
- Enuresis: take at bedtime; restrict fluids from 1 h before to 8 h after dose.
- Doses differ greatly between routes — check carefully.

## DOSAGES

- **Diabetes insipidus:** PO 100–200 mcg three times daily; intranasal 10–40 mcg daily; IV 1–4 mcg daily.
- **Enuresis:** 200–400 mcg tablet (120–240 mcg melt) at bedtime.
- **Haemophilia/vWD:** 0.3 mcg/kg IV over 20–30 min (max 20 mcg).
- **Nocturia (adults <65):** 25–50 mcg sublingual at bedtime.

## PHARMACOKINETICS

- **Onset:** PO 1 h; IV 15–30 min. **Duration:** 6–24 h.
- Resistant to enzymatic degradation; oral bioavailability ~0.1%.
- Excreted in urine. **t½:** 2–3 h.

## PHARMACODYNAMICS

- **V2-receptor** agonist in collecting duct → aquaporin insertion → ↑ water reabsorption, ↓ urine volume.
- ↑ release of factor VIII and von Willebrand factor from endothelium.
- Minimal V1 (vasopressor) activity.

## SIDE & ADVERSE EFFECTS

### SIDE EFFECTS (COMMON)

- Headache.
- Nasal congestion (spray), abdominal pain, nausea.
- Facial flushing (IV).

### ADVERSE EFFECTS (SERIOUS)

- Hyponatraemia and water intoxication → headache, vomiting, confusion, seizures (esp. children, elderly, excess fluids).
- Tachyphylaxis with repeated haemostatic doses.
- Thrombosis (rare).

## CONTRAINDICATIONS & CAUTIONS

- Hyponatraemia; habitual or psychogenic polydipsia.
- Heart failure, conditions requiring diuretics.
- SIADH; moderate–severe renal impairment (CrCl <50 for some products).
- von Willebrand type 2B.

### KEY INTERACTIONS

- **Drugs causing SIADH (SSRIs, TCAs, carbamazepine, chlorpromazine), NSAIDs:** ↑ hyponatraemia.
- **Loperamide:** ↑ desmopressin levels.

## NURSING MANAGEMENT

- Baseline and follow-up serum sodium (especially elderly, within 1 week).
- Enforce fluid restriction around dose for enuresis.
- Daily weight, fluid balance in DI.
- Stop if vomiting/diarrhoea until fluid balance normal.
- Haemostatic use: monitor BP, HR, factor levels.

## PATIENT EDUCATION

- Only sip fluids from 1 h before to 8 h after the bedtime dose.
- Stop the medicine if your child has vomiting or diarrhoea.
- Report headache, nausea, confusion or swelling (water retention).
- Use enuresis alarm and toilet routine too.

**KEY POINTS**

- V2 agonist — antidiuretic.
- Hyponatraemia; fluid restriction.
- DI, bed-wetting, nocturia, mild haemophilia/vWD.
- Route-specific doses.